

Harshvardhan Bhosale

hvbhosale99@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | New York | (716) 486-9802

EDUCATION

Master of Science in Computer Science

December 2025

University at Buffalo, Buffalo, NY

Bachelor of Science in Computer Science

June 2024

Indian Institute of Information Technology, India

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, TypeScript, Go, Rust, SQL

Backend and Distributed Systems: Spring Boot, FastAPI, Node.js, gRPC, GraphQL, WebRTC, Socket.io

Frontend: React.js, Next.js

Cloud, DevOps & Observability: AWS, Docker, Kubernetes, Jenkins, Grafana

Databases: PostgreSQL, MongoDB, Redis

EXPERIENCE

Software Development Intern | Nomura

January 2024 – July 2024

- Developed a dependency upgrade tool using TypeScript, orchestrating seamless integration via GitLab pipelines, significantly reducing manual intervention and ensuring consistent updates.
- Upgraded three Oracle database instances, achieving a 40% increase in data processing speed.
- Constructed a robust Spring Boot application utilizing JMS, JDBC, and Oracle SQL for trade reporting achieving a 25% reduction in trade processing time.
- Improved documentation accessibility by scraping Confluence pages to generate vectorized word embeddings, successfully training GPT-based language models that achieved an average BLEU score of 51%.

Software Development Intern | Microsoft (Nuance Communications)

May 2023 – July 2023

- Managed the ingestion, caching, and processing of over 1 million radiology reports using Azure Redis Cache, significantly reducing retrieval and processing latency by over 50%.
- Implemented and evaluated solutions for semantic highlighting of critical medical text, substantially enhancing the readability and accuracy of radiology reports.
- Developed NLP-driven text summarization methods, enabling rapid synthesis of medical data from detailed reports, thus accelerating clinical review processes.

PROJECTS

Mimir – Real-Time AI Voice Agents for High-Volume NGO Operations | Python, FastAPI, PostgreSQL, AWS Fargate, AWS Lambda

- Built and shipped end-to-end real-time VoIP AI agents for customer support using FastAPI, WebRTC, LangChain, and OpenAI Realtime via Twilio Media Streams, cutting average handle time by 42% and improving first-contact resolution by 28% across 15+ NGO deployments.
- Architected multi-tenant versioning with per-tenant vector stores and rate-isolated pipelines, enabling 5x concurrency scaling while maintaining sub-250 ms round-trip latency for speech-to-action loops.
- Integrated Semantic VAD and intent-aware turn-taking to optimize barge-in and silence detection, reducing interruption errors and boosting CSAT by 18 points through more natural conversational flow.
- Implemented telephonic RAG with tool-use for ticket lookups, order status, and booking flows, automating 65% of repetitive Tier-1 tasks delivering 1.3x ROI within 90 days.

Enigma – Real-Time Secure Messaging Platforms | JavaScript, MongoDB, Express.js, React.js, Node.js, Socket.io, OAuth,

- Engineered and deployed a full-stack, real-time chat application leveraging the MERN stack, integrating Socket.IO for bidirectional communication.
- Architected and implemented scalable backend APIs, intuitive React-based frontend components, and a robust MongoDB schema, ensuring seamless real-time message delivery and persistent storage.
- Boosted application performance by implementing efficient data handling and connection management with Socket.IO, capable of supporting concurrent users without degradation in real-time responsiveness.

DiveDash – D&I Monitoring Tool | Java, Spring Boot, PostgreSQL, React, DaisyUI, AWS, Kubernetes, Grafana

- Engineered and deployed a containerized Spring Boot microservices backend on Kubernetes (K8s) for scalable orchestration and high availability; integrated Grafana for real-time observability and performance tuning.
- Implemented an enterprise-grade security layer using Spring Security (OAuth2/JWT) for API authentication and enforced strict data consistency via ATOMIC transactions in PostgreSQL to manage highly concurrent D&I metric processing.